Localized in-gap eigenstates of a 10^4-vertex disordered photonic network from direct interior Maxwell eigensolves

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Bottom-up (611 bands) and interior (bandpass Chebyshev filtered subspace iteration; 133 certified pairs near eigenvalue index 5000) Maxwell eigensolves on a 12 GB GPU, validated against MPB, resolve the gap-edge modes of local-self-uniform photonic networks with 10^3 and 10^4 vertices. The 10^3 gap is empty; the 10^4 pseudogap holds five certified states with $\xi = 1.8$–$2.5\,\mu$m $\ll L = 24.6\,\mu$m, robust to re-rasterization and refinement. Decay lengths funnel to $\sim 2\,\mu$m at the edges, size-independent between boxes; level statistics are GOE-like in the bands and Poisson-compatible at the lower edge. No mobility edge is claimed.

Motivation.—Local-self-uniform (LSU) networks are disordered trivalent dielectric networks that open a full three-dimensional photonic band gap without long-range order [1], in the family of amorphous-diamond [2, 3] and hyperuniform [4–9] photonic materials. Gap formation and Anderson localization [10, 11] in such correlated disordered media are linked [12, 13], and a transition from diffusion to localization near the band edge has been observed in 3-D amorphous networks [14], but what the eigenmodes *inside* and at the edges of the gap look like at the 10^4-vertex scale has not been resolved by direct eigenmode computation. Sellers *et al.* state that an N-vertex supercell has its gap between bands $N/2$ and $N/2+1$ [1], consistent with the band count of the parent gyroid/srs crystal [15], so at $N = 10^4$ the gap sits at eigenvalue index ≈ 5000; computing all lower bands with the validated bottom-up solver would need 1.38 TB of locked vectors at 256^3 and $O(100)$ days—infeasible—which is why an interior method was built.

System and operator.—We use the $N = 1000$ reference network of Ref. [1] (box $L = 11.44\,\mu$m, mean rod length $0.80\,\mu$m) and an $N = 10^4$ network generated with the same protocol at the same vertex density ($0.668\,\mu\text{m}^{-3}$, $L = 10^{1/3} \times 11.44 = 24.647\,\mu$m). Rods are rasterized as binary voxels (no subpixel smoothing [16, 17]) in two decorations: the "elliptical" convention of the original $N = 1000$ montage (circular radius $0.2252\,\mu$m in a z-compressed space stretched by 2.5, $n = 2.9275$, $\varepsilon = 8.57$, filling fraction ff $= 21.7\%$ at 128^3 and 256^3), and a "circular" decoration (radius $0.331836\,\mu$m, $n = 2.9$, $\varepsilon = 8.41$) whose radius was bisected to ff $= 22.00\%$ at 256^3 (22.01% at 192^3; 21.91% at 128^3 for $N = 1000$). Following MPB [18] we solve $\Theta \boldsymbol{H} = \nabla \times \varepsilon^{-1} \nabla \times \boldsymbol{H} = (\omega/c)^2 \boldsymbol{H}$ in the two-component transverse plane-wave basis of the periodic supercell at $k = \Gamma$; one application costs six 3-D FFTs, the two $\omega = 0$ modes are removed exactly, and Θ is real symmetric on real fields. Eigenvalues are quoted as $\lambda = (\omega/c)^2$ in μm^{-2} and as $\nu = \omega a/2\pi c$ with $a = L/5 = 2.288\,\mu$m (the density-equivalent 8-vertex cell; identical for both sizes). Band numbering is MPB's (bands 1–2 are the $\omega = 0$ pair).

Method.—(a) For $N = 1000$ we use a deflated block LOBPCG [19, 20] with MPB's transverse-projection preconditioner [18], blocks of 32 with 12 guard vectors, complex64 vectors and operator on the GPU, all Gram/Rayleigh–Ritz algebra in fp64 on the host (TF32 matmuls disabled), per-band relative residual $\leq 10^{-4}$, and a host-streamed locked set. On identical 64^3 grids it reproduces MPB (fp64, tolerance 10^{-7}) band by band: max $\Delta\omega/\omega = 9.0 \times 10^{-6}$ over 298 bands and 3.5×10^{-5} over 658 bands (medians 1.4×10^{-6}); complex64 vs complex128 differ by $\leq 2.4 \times 10^{-7}$ over 96 bands. The pre-registered convergence gate failed as registered: over $64^3/96^3/128^3$ five of six sampled bands converge monotonically with Richardson residual $\leq 0.23\%$ at 128^3, but the gap-edge band 500 is non-monotone with a 0.27% spread (1800× the solver–MPB error: rasterization-limited), and the elliptical gap width is non-monotone, $2.35 \to 1.93 \to 2.08\%$. (b) For $N = 10^4$ we first locate the gap with the kernel polynomial method [21, 22] (KPM; 256^3, degree 12 000, 12 probes, Jackson kernel, $\lambda_{\max} = 3975$), then solve the window $\lambda \in [1.757, 2.117]$ at 192^3 by two-stage bandpass Chebyshev filtered subspace iteration [23–26]: a Jackson step-difference filter of degree 3000–4000 builds a basis of 1.5× the predicted count from random starts and the same filter at degree 12 000 polishes it; candidates are extracted only by fp64 Rayleigh–Ritz on the original Θ and accepted only if their residual on Θ is $\leq 10^{-4}$, so ghosts are excluded by construction (none were observed). The method won a measured bake-off on a 50-band interior slice of the exact $N = 1000$ spectrum against folded-spectrum LOBPCG [27, 28] and shift-invert with preconditioned MINRES; filtered Lanczos [29] was excluded on memory grounds [Supplemental Material (SM) [30]]. Its accuracy is set by re-solving that slice in the production configuration: 50/50 found, 0 ghosts, max $\Delta\lambda/\lambda = 2.4 \times 10^{-7}$ (median 6×10^{-8})—after correcting a Rayleigh-quotient normalization bug that biased every eigenvalue high by $+4.7 \times 10^{-5}$ ($\lambda_{\text{corr}} = \lambda_{\text{raw}}/\|x\|^2$, applied retroactively; SM)—and by an independent cross-check on the circular $N = 1000$ decoration (210/210 states, max 4.1×10^{-7}).

Localization is quantified per mode from $u = \varepsilon|\boldsymbol{E}|^2$ by the participation fraction $p = (\sum u)^2/(N_{\rm vox}\sum u^2)$ and an envelope decay length ξ from a linear fit of $\ln u(r)$ vs the minimum-image distance r from the peak over $[0.10, 0.95]\,L/2$, $u \propto e^{-2r/\xi}$; a fit is "unresolved" (lower bound only) if $\xi \geq L/2$, if u decays by less than one decade, or if $r^2 < 0.7$ (all per-mode values: SM tables).

Results, $N = 1000$.—All 611 bands (through MPB band 613) of the elliptical structure at 128^3 were solved in 5.5 h. The gap lies between bands 500 and 501 as preregistered: $\lambda = 1.8830 \to 1.9628\,\mu\mathrm{m}^{-2}$, $\Delta\nu/\nu = 2.08\%$ at 128^3 (centre $\nu = 0.505$), a spacing $9\times$ the median of its 20 neighbours. With the circular decoration the gap is again at 500|501 and widens to $1.8276 \to 2.0225\,\mu\mathrm{m}^{-2}$, $\Delta\nu/\nu = 5.07\%$ ($25\times$ the neighbouring spacings). Both gaps are empty of eigenvalues. The regenerated 210-tile montage of bands 398–607 reproduces the original cluster montage in character (SM). Band-edge modes are localized and deep-window modes extended [Fig. 2(a)]: for the circular decoration $\xi = 1.47\,\mu\mathrm{m}$ (band 500, $r^2 = 0.97$, $p = 0.42\%$) and $2.30\,\mu\mathrm{m}$ (band 501, $p = 3.9\%$), vs $p = 12$–37% far from the gap; for the elliptical decoration $\xi = 1.82$ and $2.92\,\mu\mathrm{m}$. Only 42 of 210 circular-decoration modes have resolved ξ; 168 hit the $L/2 = 5.72\,\mu\mathrm{m}$ ceiling or fail the exponential-shape test, which is the quantitative reason for the $N = 10^4$ computation.

Results, $N = 10^4$.—KPM places 5010.3 ± 12.5 transverse states below $\lambda = 1.957$, i.e. 0.501 bands per vertex, confirming the $N/2$ rule at this scale (gap at MPB bands $\approx 5012|5013 \pm 13$). The Jackson-smoothed KPM density falls below 160 states per unit λ over $[1.864, 1.996]\,\mu\mathrm{m}^{-2}$ ($\Delta\nu/\nu = 3.4\%$; the 5% and 20% criteria of the project give $[1.885, 1.968]$ and $[1.837, 2.022]$, a ± 0.03 criterion systematic), narrower than the 5.07% gap of the $N = 1000$ box with the same decoration and contained within it. Three overlapping slices at 192^3 ($69 + 5 + 61$ pairs; two states found twice were de-duplicated by eigenvector overlap 0.9988, their corrected eigenvalues agreeing to $\leq 1.2 \times 10^{-8}$) give **133 residual-certified eigenpairs** in $[1.75705, 2.11657]\,\mu\mathrm{m}^{-2}$ (worst residual 5.9×10^{-5}; 104 GPU-hours, 1.16×10^7 operator applications), with no in-window pair left unconverged. Completeness is certified on the sub-interval $[1.9063, 1.9606]$ by a deflated-probe KPM count of missed states, -0.0002 ± 0.0001; on the full window the same estimator ($+1.7 \pm 0.3$) has an edge-leakage bias of at least $+1.2 \pm 0.2$ and is a consistency check only (SM). The largest interior spacing of the certified spectrum runs from 1.8860 to $1.9264\,\mu\mathrm{m}^{-2}$ ($\Delta\nu/\nu = 1.06\%$, $28\times$ the median spacing); but the nominal gap is not empty. **Ten certified states lie inside** $[1.864, 1.996]$, at $\lambda = 1.8690$, 1.8708, 1.8731, 1.8860, 1.9264, 1.9296, 1.9441, 1.9472, 1.9738, 1.9901 [Fig. 1(d), Table SI]. Four of them are artefacts of the rasterization convention: rods whose radius pokes through a box face are not wrapped, so the outermost voxel shell has ff $= 0.1975$ against 0.2211 in the interior (an 11% deficit forming a planar defect), and the states at 1.8708, 1.8731, 1.9296, 1.9472 carry 18, 24, 44 and 42% of their energy in that shell (6.1% of the volume; three peak on a box edge). Re-solving the gap window with minimum-image wrapping (0.078% of voxels change) finds no partner for three of them (best overlaps 0.006, 0.131, 0.090); the fourth (1.9472, overlap 0.30) is left undetermined because the periodic solve is itself measured incomplete ($+2.0 \pm 0.4$ missed), so absence of a partner is not proof of absence. A fifth in-gap state (1.9441) is extended, not localized ($\xi = 13.0\,\mu\mathrm{m} > L/2$, $r^2 = 0.33$, $p = 0.56\%$). **Five bulk-localized in-gap states remain**: $\xi = 1.80$–$2.51\,\mu\mathrm{m}$ ($\xi/L = 0.07$–0.10), $p = 0.034$–0.33% (participation volumes 5–49 $\mu\mathrm{m}^3$ in a 15 000 $\mu\mathrm{m}^3$ box), $r^2 = 0.971$–0.998, shell energy 2.5–10.5%; all five persist under periodic wrapping with overlaps 0.994–1.000 and shifts $\Delta\lambda = -0.0007$ to -0.0034, every shift negative as required when dielectric is added. Their identity survives a $2.4\times$ voxel refinement: at 256^3 the same 11 states are found in $[1.84, 1.95]$, matched one-to-one by eigenvector overlap (min 0.873, median 0.996), with $|\Delta\omega/\omega| \leq 0.34\%$ (low edge) and $\leq 0.04\%$ (high edge; 3 of 11 fine pairs certified). That is good evidence against a rasterization artefact, but with $192^3 = 7.8$ voxels/μm and the 0.27% gap-edge spread seen at $N = 1000$, sub-percent gap-edge frequencies are not claimed.

The decay length varies smoothly across the window [Fig. 3(a)]: 121 of 133 fits are resolved (12 fail $r^2 \geq 0.7$, one of which also exceeds the ceiling), and the medians by spectral bin are $5.0 \to 3.0 \to 2.1\,\mu\mathrm{m}$ approaching the lower edge, $2.1\,\mu\mathrm{m}$ inside, and $2.8 \to 4.7 \to 6.1\,\mu\mathrm{m}$ climbing away above; the participation fraction spans three decades, 0.03%–12% [Fig. 3(b)], and the participation-volume radius $(3V_p/4\pi)^{1/3}$ follows the same funnel ($2.7 \to 1.5 \to 1.2\,\mu\mathrm{m}$ and $1.4 \to 3.1 \to 4.1 \to 5.4\,\mu\mathrm{m}$). The envelope fit is not a precision quantity: shrinking the fit range to $[0.10, 0.60]\,L/2$ moves ξ by a median 30% and the window-end/near-edge contrast from 1.7–2.2$\times$ to 1.6–1.8$\times$; the funnel is robust, three-figure ξ values are not.

Can we claim Anderson localization?—We take the items in turn. (1) *Spatial localization*: supported. The in-gap and gap-edge states have $\xi = 1.8$–$2.5\,\mu\mathrm{m}$ in $L = 24.6\,\mu\mathrm{m}$, $p \leq 0.33\%$ and $r^2 \geq 0.97$; the tiles [Fig. 2(b)] show single compact blobs a few μm across. ξ is an envelope-fit length with tens-of-percent fit-range sensitivity, not a transport length. (2) *Anderson vs rare-region (Lifshitz-tail [31]) states*: not separable with this data. Both boxes are single realizations with the same local statistics (filling fraction coarse-grained at $2\,\mu$m: mean 0.2204 vs 0.2209, s.d. 0.0398 vs 0.0405), but $N = 10^4$ has ten times as many independent ξ-cells and samples deeper tails; at 5 in-gap states per $N = 10^4$ volume an empty $N = 1000$ gap has probability $e^{-0.5} = 0.61$. A direct test (energy-weighted local filling fraction of candidates vs bulk controls) was run and *retracted*: it was not

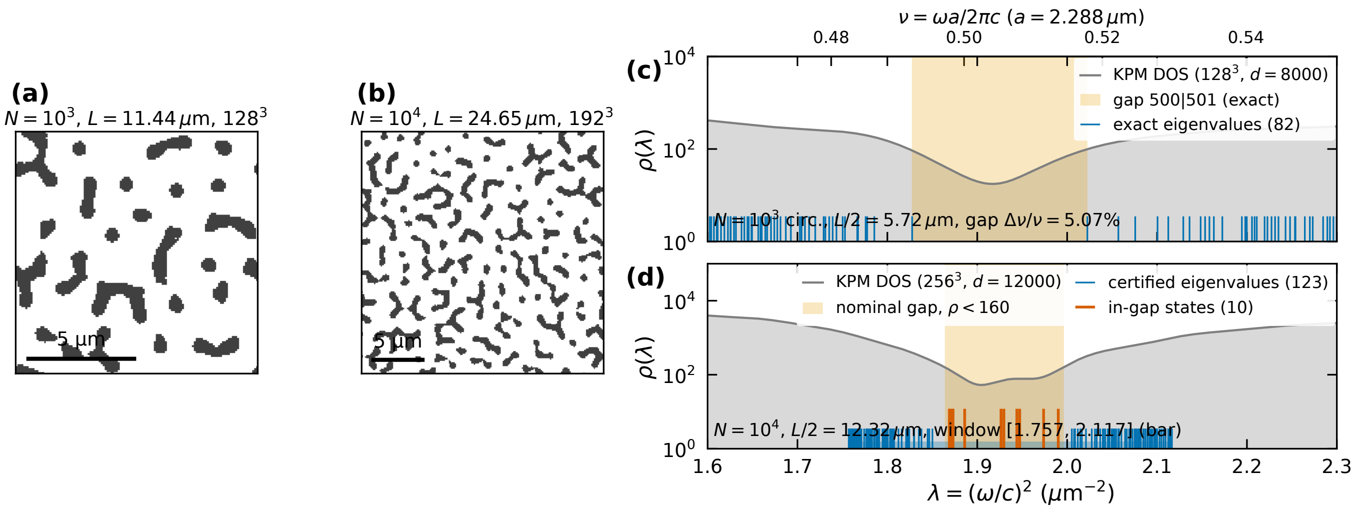

FIG. 1. (a,b) Mid-plane $\varepsilon(\boldsymbol{r})$ of the $N = 10^3$ (128^3) and $N = 10^4$ (192^3) networks, circular decoration (dark = dielectric). (c) $N = 10^3$: exact eigenvalues (bottom-up, 611 bands) around the gap, on the KPM density of states of the same structure; the shaded band is the exact gap between MPB bands 500 and 501. (d) $N = 10^4$: KPM density of states (256^3) with the 133 residual-certified 192^3 eigenvalues (blue) and the ten inside the nominal gap $\rho < 160$ states per unit λ (red); bar: solved window. Top axis: $\nu = \omega a/2\pi c$, $a = 2.288\,\mu$m.

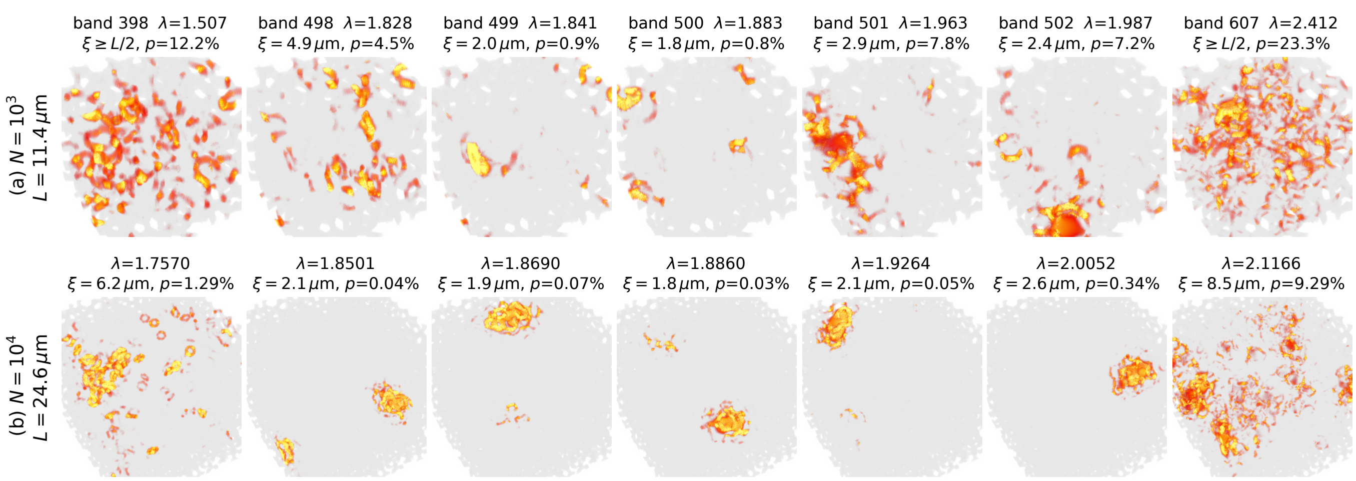

FIG. 2. Energy density $\varepsilon|\boldsymbol{E}|^2$ (volume render, per-tile colour scale clipped at the 99.9th percentile; network in translucent grey). (a) $N = 10^3$, elliptical decoration, bands 398, 498–502, 607 around the 500|501 gap. (b) $N = 10^4$, circular decoration: window bottom, lower gap edge, three in-gap candidates, upper gap edge, window top. ξ and p as defined in the text; "$\xi \geq L/2$" marks a ceiling-limited fit.

matched in mode extent, and against an extent-matched translation null the ordering reverses (SM). A rare-region origin is therefore the leading, not the established, explanation. (3) *Finite-size scaling*: two sizes only. Matched-decoration band-edge states have $\xi = 1.47$ and $2.30\,\mu$m at $L = 11.44\,\mu$m and 1.80 and $2.09\,\mu$m at $L = 24.65\,\mu$m [edge states 1.8860 and 1.9264; Fig. 4(b)], with participation volumes 6 and $58\,\mu\text{m}^3$ vs 5 and $7\,\mu\text{m}^3$: ξ is L-independent within the fit uncertainty, consistent with genuine localization rather than a finite-size effect, but two points are not a scaling analysis and 168 of 210 $N = 1000$ modes are ceiling-limited. (4) *Level statistics* [32, 33]: Θ at Γ with real ε is real symmetric, so the Wigner–Dyson class is orthogonal and the references are $\langle r\rangle_{\text{P}} = 2\ln 2 - 1 = 0.386$ and $\langle r\rangle_{\text{GOE}} = 0.531$ [34, 35]. The adjacent-gap ratio needs no unfolding and is computed within contiguous bands [Fig. 4(a)]. In the extended regions it is GOE-like: 0.528 ± 0.038 (51 levels, $\lambda < 1.816$) and 0.528 ± 0.041 (42 levels, $\lambda > 2.051$) at $N = 10^4$, pooled 0.519 ± 0.024 outside the bracket; 0.508 ± 0.010 and 0.533 ± 0.011 over all 611 $N = 1000$ levels (elliptical, circular). In the 12 levels just below the gap ($1.821 \leq \lambda \leq 1.850$), where ξ has dropped to 2–3 μm, $\langle r\rangle = 0.357 \pm 0.073$, 1.7σ below GOE and con-

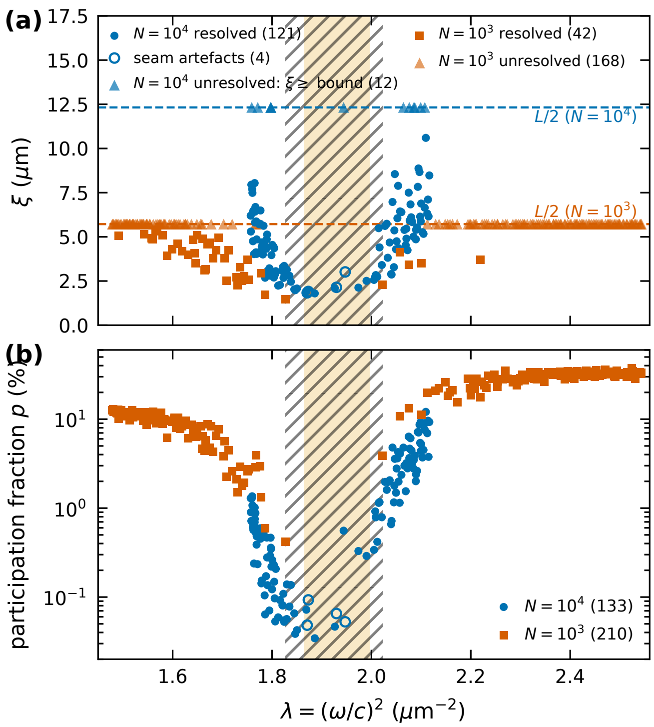

FIG. 3. (a) Envelope decay length ξ vs λ for $N = 10^4$ (circles, 192^3) and $N = 10^3$ circular decoration (squares, 128^3); dashed lines are the finite-size ceilings $L/2$; triangles at the ceiling are unresolved fits (lower bounds); open circles are the four seam artefacts; shaded: nominal $N = 10^4$ gap; hatched: $N = 10^3$ gap. (b) Participation fraction. Same decoration for both sizes.

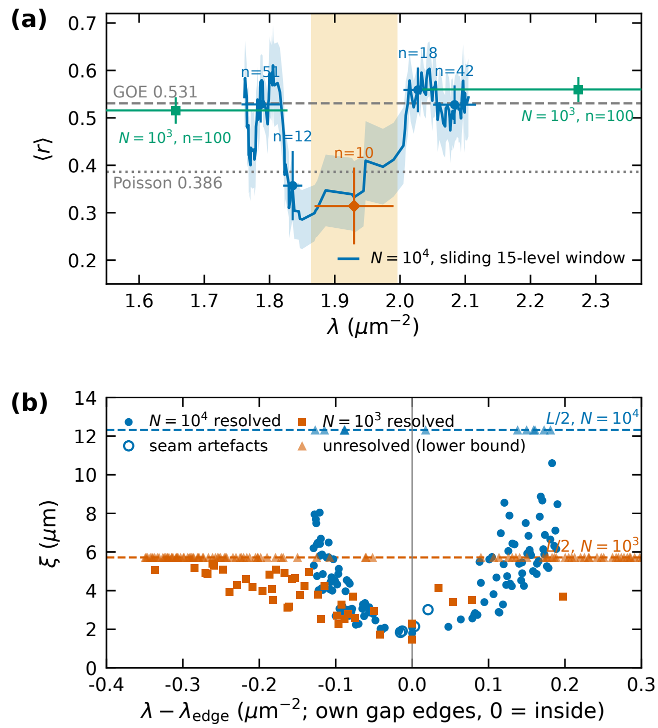

FIG. 4. (a) Adjacent-gap ratio $\langle r \rangle$: $N = 10^4$ sliding 15-level window (line, ± 1 s.e.) and contiguous bands (circles: 51, 12, 18, 42 levels; diamond: ten in-gap levels; bootstrap errors); $N = 10^3$ circular, bands 401–500 and 501–600 (squares). Dashed/dotted: GOE 0.531 / Poisson 0.386. (b) Matched-decoration comparison: resolved ξ vs distance from each box's own gap edges ($N = 10^3$: 1.8276|2.0225; $N = 10^4$: 1.8860|1.9264; in-gap states at 0).

sistent with Poisson; above the gap the 18 nearest levels (2.005–2.050, $p \approx 1\%$) give 0.559 ± 0.047. The ten in-gap levels give 0.31 ± 0.08, but ten levels, four of them seam artefacts, are not a statistic. Uncertainties are bootstrap; the eigenvalue accuracy ($\Delta\lambda/\lambda \sim 2 \times 10^{-7}$) is three orders below the median spacing (1.4×10^{-3}). The trend is the expected one—level repulsion where states are extended, its loss where they are localized—but the near-edge sample is small: suggestive, not conclusive. A Thouless-type sensitivity to boundary twists cannot be obtained from the saved Γ-point data. (5) *Mobility edge*: not claimable. No transport quantity was computed, there is one realization per size, and no scaling; the $\xi(\omega)$ funnel of Fig. 3 and the $\langle r \rangle$ drop of Fig. 4(a) are the *phenomenology* of a localization–delocalization crossover at the gap edge, the regime probed experimentally in 3-D amorphous networks [14] and predicted for correlated disorder [13]. (6) *Vector-wave caveat*: early reports of 3-D localization of light [36] were not confirmed, localization is absent in 3-D ensembles of uncorrelated point scatterers when the vector nature of light is kept [37–39], and it has been demonstrated numerically only for random aggregates of overlapping metallic spheres, not for dielectric ones [40]; localization at the edge of a photonic gap in a correlated network [12, 13, 41] is a different mechanism, and it is the one this data speaks to.

Verdict.—The defensible claim is: strongly spatially localized in-gap and gap-edge eigenstates with $\xi \ll L$, size-independent between $L = 11.4$ and $24.6\,\mu$m, consistent with Anderson-type localization at the edges of a photonic pseudogap in a correlated disordered network, accompanied by GOE-like level statistics in the extended bands and a Poisson-compatible ratio in the 12 levels nearest the lower edge. The claim of an Anderson transition or a mobility edge requires finite-size scaling over three or more sizes and/or transport or larger-sample level statistics that this data does not contain; the level statistics computed here are suggestive of the expected crossover but inconclusive at the gap edge.

Limitations.—Gap-edge frequencies are rasterization-limited (0.27% non-monotone spread at $N = 1000$; $\leq 0.34\%$ between 192^3 and 256^3 at $N = 10^4$), and on MPB's interpolated read-back of the 64^3 grid the elliptical 500|501 spacing shrinks to 0.60% with the largest spacing at 501|502, so the gap position at 64^3 is a statement about the binary-voxel convention. Decay lengths are bounded by $L/2$ and fit-range dependent at the tens-of-percent level. There is one disorder realization per size. Arithmetic is fp32 on the device with fp64 Rayleigh–Ritz; the merged 133-mode set is not mutually orthonormal to the registered gate ($\|G - I\|_{\max} = 4.9 \times 10^{-4}$ across slices vs 8.6×10^{-6} within), which affects no spectrum, montage or localization result. Four claims made during the project were withdrawn and are absent here: the rare-region z-test, the statement that the resolution confound is "closed" (its 99.98% power-retention statistic is saturated: a 96^3 grid retains 99.8%), the "vanished" verdict for the state at 1.9472, and all uncorrected eigenvalues. Full records, the gate ledger and every number with its source file are in the SM [30].

[Acknowledgments to be filled.]

[1] S. R. Sellers, W. Man, S. Sahba, and M. Florescu, Nature Communications **8**, 14439 (2017).
[2] K. Edagawa, S. Kanoko, and M. Notomi, Physical Review Letters **100**, 013901 (2008).
[3] S. Imagawa, K. Edagawa, K. Morita, T. Niino, Y. Kagawa, and M. Notomi, Physical Review B **82**, 115116 (2010).
[4] M. Florescu, S. Torquato, and P. J. Steinhardt, Proceedings of the National Academy of Sciences **106**, 20658 (2009).
[5] W. Man, M. Florescu, E. P. Williamson, Y. He, S. R. Hashemizad, B. Y. C. Leung, D. R. Liner, S. Torquato, P. M. Chaikin, and P. J. Steinhardt, Proceedings of the National Academy of Sciences **110**, 15886 (2013).
[6] S. F. Liew, J.-K. Yang, H. Noh, C. F. Schreck, E. R. Dufresne, C. S. O'Hern, and H. Cao, Physical Review A **84**, 063818 (2011).
[7] N. Muller, J. Haberko, C. Marichy, and F. Scheffold, Advanced Optical Materials **2**, 115 (2014).
[8] N. Muller, J. Haberko, C. Marichy, and F. Scheffold, Optica **4**, 361 (2017).
[9] M. A. Klatt, P. J. Steinhardt, and S. Torquato, Proceedings of the National Academy of Sciences **116**, 23480 (2019).
[10] P. W. Anderson, Physical Review **109**, 1492 (1958).
[11] E. Abrahams, P. W. Anderson, D. C. Licciardello, and T. V. Ramakrishnan, Physical Review Letters **42**, 673 (1979).
[12] S. John, Physical Review Letters **58**, 2486 (1987).
[13] L. S. Froufe-Pérez, M. Engel, J. J. Sáenz, and F. Scheffold, Proceedings of the National Academy of Sciences **114**, 9570 (2017).
[14] J. Haberko, L. S. Froufe-Pérez, and F. Scheffold, Nature Communications **11**, 4867 (2020).
[15] L. Lu, L. Fu, J. D. Joannopoulos, and M. Soljačić, Nature Photonics **7**, 294 (2013).
[16] A. Farjadpour, D. Roundy, A. Rodriguez, M. Ibanescu, P. Bermel, J. D. Joannopoulos, S. G. Johnson, and G. W. Burr, Optics Letters **31**, 2972 (2006).
[17] C. Kottke, A. Farjadpour, and S. G. Johnson, Physical Review E **77**, 036611 (2008).
[18] S. G. Johnson and J. D. Joannopoulos, Optics Express **8**, 173 (2001).
[19] A. V. Knyazev, SIAM Journal on Scientific Computing **23**, 517 (2001).
[20] J. A. Duersch, M. Shao, C. Yang, and M. Gu, SIAM Journal on Scientific Computing **40**, C655 (2018).
[21] A. Weiße, G. Wellein, A. Alvermann, and H. Fehske, Reviews of Modern Physics **78**, 275 (2006).
[22] L. Lin, Y. Saad, and C. Yang, SIAM Review **58**, 34 (2016).
[23] Y. Zhou, Y. Saad, M. L. Tiago, and J. R. Chelikowsky, Journal of Computational Physics **219**, 172 (2006).
[24] A. Pieper, M. Kreutzer, A. Alvermann, M. Galgon, H. Fehske, G. Hager, B. Lang, and G. Wellein, Journal of Computational Physics **325**, 226 (2016).
[25] R. Li, Y. Xi, L. Erlandson, and Y. Saad, SIAM Journal on Scientific Computing **41**, C393 (2019).
[26] J. Winkelmann, P. Springer, and E. Di Napoli, ACM Transactions on Mathematical Software **45**, 21 (2019).
[27] L.-W. Wang and A. Zunger, The Journal of Chemical Physics **100**, 2394 (1994).
[28] C. Vömel, S. Z. Tomov, O. A. Marques, A. Canning, L.-W. Wang, and J. J. Dongarra, Journal of Computational Physics **227**, 7113 (2008).
[29] H.-r. Fang and Y. Saad, SIAM Journal on Scientific Computing **34**, A2220 (2012).
[30] See Supplemental Material at [URL will be inserted by publisher] for the complete technical record: structure generation and rasterization (including the boundary-seam artefact), operator and solver details, all pre-registered gates with outcomes, the bake-off, the Rayleigh-normalization correction, completeness checks, localization methodology, level-statistics details, the full 133-state table, retracted claims and the full list of limitations.
[31] I. M. Lifshitz, Advances in Physics **13**, 483 (1964).
[32] B. I. Shklovskii, B. Shapiro, B. R. Sears, P. Lambrianides, and H. B. Shore, Physical Review B **47**, 11487 (1993).
[33] F. Evers and A. D. Mirlin, Reviews of Modern Physics **80**, 1355 (2008).
[34] V. Oganesyan and D. A. Huse, Physical Review B **75**, 155111 (2007).
[35] Y. Y. Atas, E. Bogomolny, O. Giraud, and G. Roux, Physical Review Letters **110**, 084101 (2013).
[36] D. S. Wiersma, P. Bartolini, A. Lagendijk, and R. Righini, Nature **390**, 671 (1997).
[37] S. E. Skipetrov and I. M. Sokolov, Physical Review Letters **112**, 023905 (2014).
[38] T. Sperling, L. Schertel, M. Ackermann, G. J. Aubry, C. M. Aegerter, and G. Maret, New Journal of Physics **18**, 013039 (2016).
[39] S. E. Skipetrov and J. H. Page, New Journal of Physics **18**, 021001 (2016).
[40] A. Yamilov, S. E. Skipetrov, T. W. Hughes, M. Minkov, Z. Yu, and H. Cao, Nature Physics **19**, 1308 (2023).
[41] S. John, Physical Review Letters **53**, 2169 (1984).